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SLOT: A

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SUBJECT: DEEP LEARNING

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Bias–Variance, Overfitting and Underfitting Analysis

SCENARIO

A student-performance prediction model uses a linear regression algorithm on nonlinear data. Compare training and testing errors to identify whether the model is underfitting.

AIM

To analyze bias, variance, overfitting, and underfitting in a student-performance prediction model using linear regression by comparing training and testing errors.

ALGORITHM

1. Import the required Python libraries.
2. Generate nonlinear student-performance data.
3. Split the dataset into training and testing sets.
4. Train a linear regression model.
5. Calculate training and testing errors.
6. Display the summary table and plots.
7. Identify underfitting by analyzing the errors.

SOURCECODE

```
import numpy as np
import pandas as pd
import matplotlib.pyplot as plt
from sklearn.model_selection import train_test_split
from sklearn.linear_model import LinearRegression
from sklearn.metrics import mean_squared_error


# Generate nonlinear student-performance data
```

```
np.random.seed(42)
X = np.linspace(1, 10, 100).reshape(-1, 1)
y = 5 * X.ravel()**2 + np.random.normal(0, 15, 100)

# Split data into training and testing sets
X_train, X_test, y_train, y_test = train_test_split(
    X, y, test_size=0.2, random_state=42
)

# Train linear regression model
model = LinearRegression()
model.fit(X_train, y_train)

# Predict training and testing outputs
train_pred = model.predict(X_train)
test_pred = model.predict(X_test)

# Calculate errors
train_error = mean_squared_error(y_train, train_pred)
test_error = mean_squared_error(y_test, test_pred)

# Create summary table
summary = pd.DataFrame({
    'Dataset': ['Training', 'Testing'],
    'Mean Squared Error': [train_error, test_error]
})

print("ERROR SUMMARY TABLE")
display(summary.round(2))
```

```
print("Training Error:", round(train_error, 2))
print("Testing Error:", round(test_error, 2))

# Plot nonlinear data and linear regression
plt.figure(figsize=(8, 5))
plt.scatter(X_train, y_train, label='Training Data')
plt.scatter(X_test, y_test, label='Testing Data')
plt.plot(X, model.predict(X), label='Linear Regression')
plt.xlabel('Study Hours')
plt.ylabel('Student Performance Score')
plt.title('Linear Regression on Nonlinear Data')
plt.legend()
plt.grid(alpha=0.3)
plt.show()

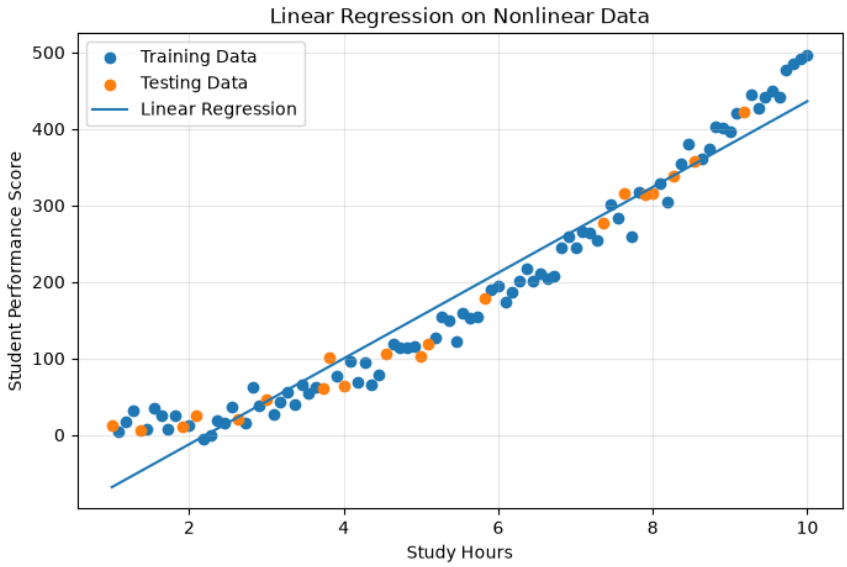
# Compare training and testing errors
plt.figure(figsize=(7, 4))
plt.bar(summary['Dataset'], summary['Mean Squared Error'],
    edgecolor='black')
plt.ylabel('Mean Squared Error')
plt.title('Training vs Testing Error')
plt.grid(axis='y', alpha=0.3)
plt.show()
```

OUTPUT

ERROR SUMMARY TABLE

	Dataset	Mean Squared Error
0	Training	1228.97
1	Testing	1022.63

```
Training Error: 1228.97
Testing Error: 1022.63
```





RESULT

Thus, training and testing errors were calculated and visualized. Since linear regression cannot adequately represent the nonlinear relationship, the model is expected to show relatively high errors on both datasets, indicating underfitting (high bias).